# Towards Worst-case Hardness for Low-Noise LPN

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Divesh Aggarwal, Rishav Gupta, Hai Hoang Nguyen,

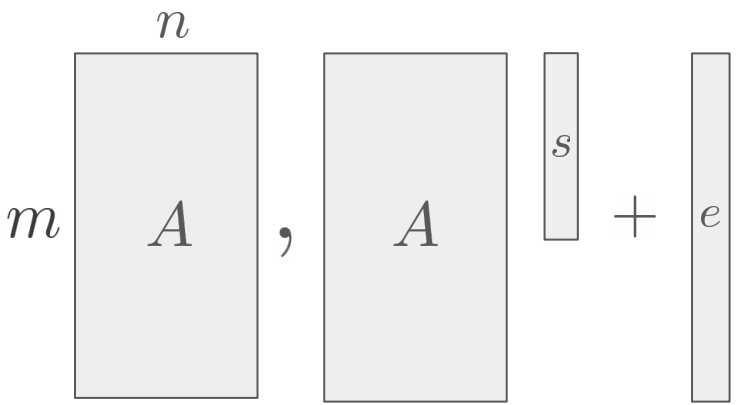
Kel Zin Tan, Prashant Nalini Vasudevan

**National University of Singapore**

# Learning Parity with Noise (LPN)

**Problem Instance:**

- Random Matrix $A \leftarrow \mathbb{F}_2^{m \times n}$
- Random secret $s \leftarrow \mathbb{F}_2^n$
- Random Noise $e \leftarrow Ber(\eta)^m$




**Search Problem**

Given $(A, As + e)$, recover $s$

**Decision Problem**

Distinguish $(A, As + e)$, from $(A, Ber(1/2)^m)$

# Hardness of LPN

Suppose m = poly(n), solve with non-negl prob

Information Set Decoding: $2^{O(\eta n)}$

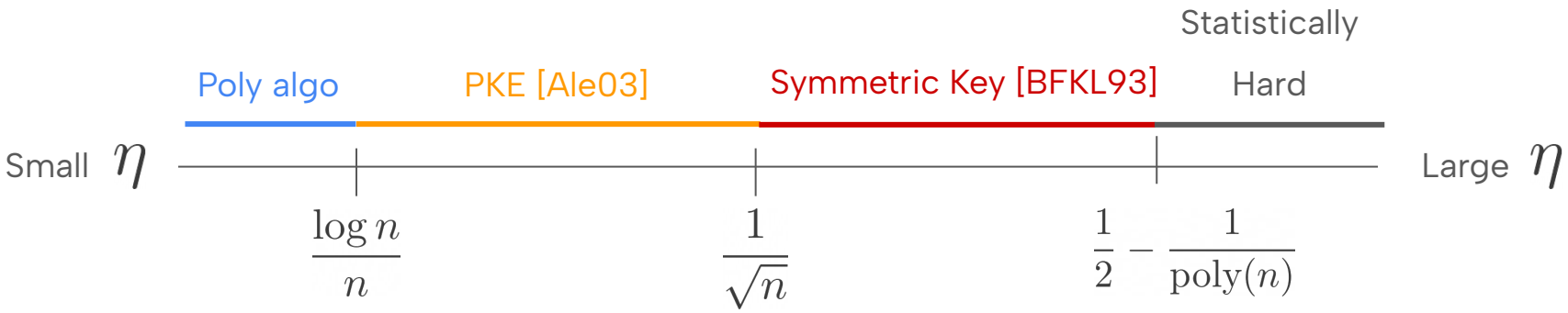
[Pra62, Ste89, Dum91, MMT11, BJMM12, MO15, BM17, BM18, DEEK24]

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# Worst-Case Version

**Nearest Codeword Problem (NCP):**

Arbitrary Matrix A; error e; secret s

Given $(A, As + e)$, find $s'$ that minimizes $|As + e - As'|$

# Worst-Case Version

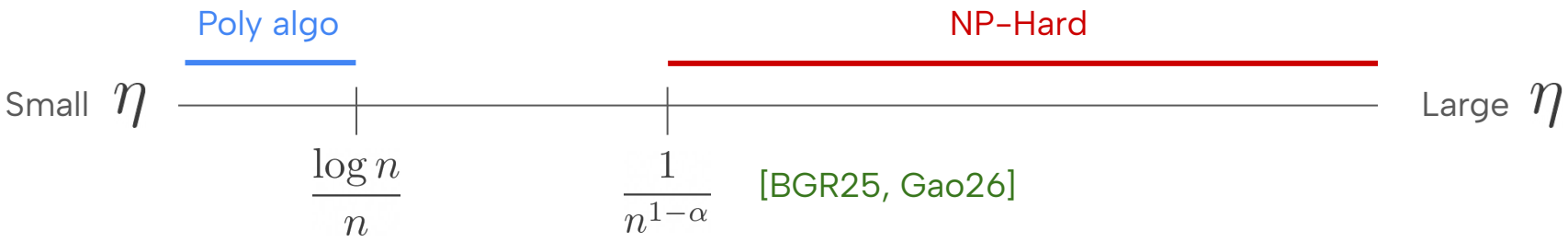
**Nearest Codeword Problem (NCP):**

Arbitrary Matrix A; error e; secret s

Given $(A, As + e)$, find $s'$ that minimizes $|As + e - As'|$

**Hardness**

Information Set Decoding: $2^{O(\eta n)}$



# Worst-Case-to-Average-Case 1

Establish average-case hardness (LPN) from worst-case (NCP)

[BLVW19, YZ21, DDRT23, DR25, PB25]

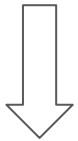
# Worst-Case-to-Average-Case 1

Establish average-case hardness (LPN) from worst-case (NCP)

[BLVW19, YZ21, DDRT23, DR25, PB25]

If Algorithm for LPN with $\eta = \frac{1}{2} - \frac{1}{\mathrm{poly}(n)}$

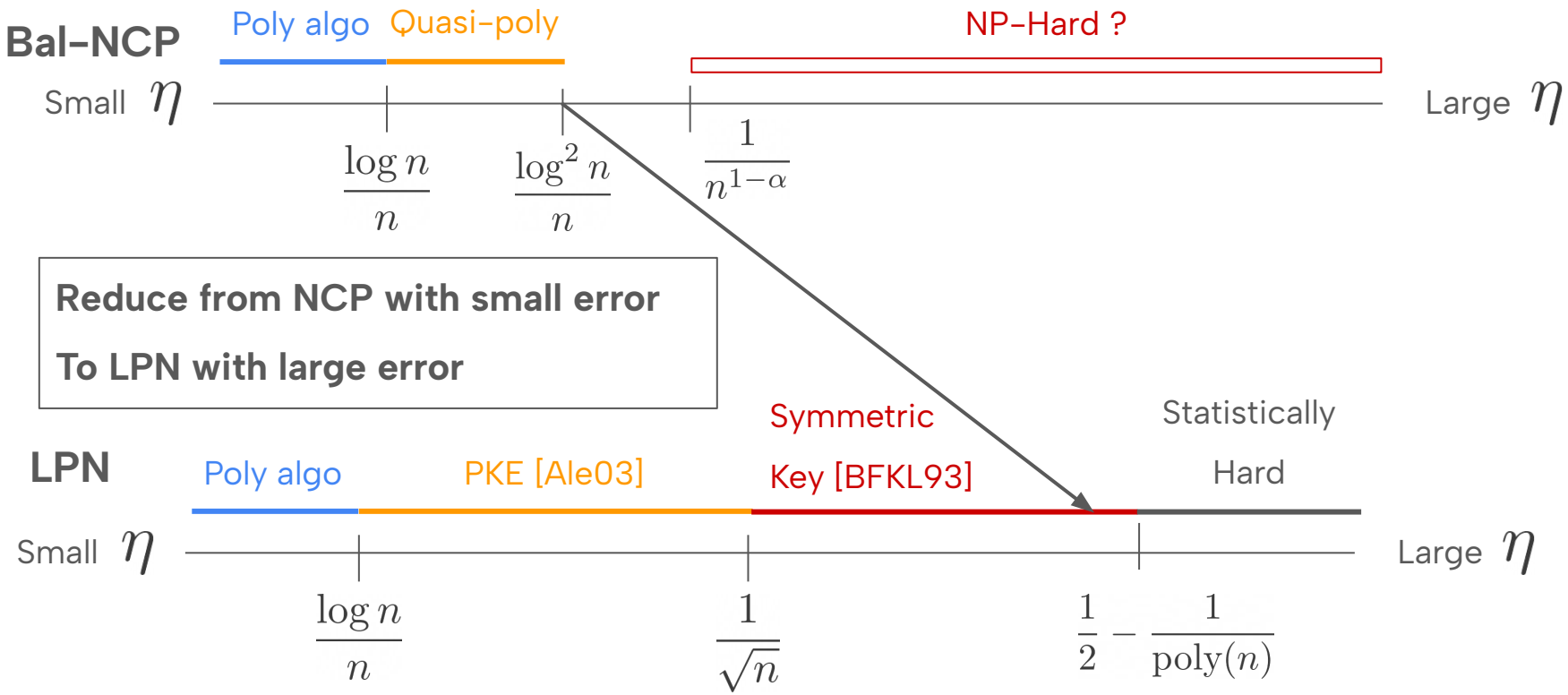
*Polynomial Reduction*



Then algorithm for Bal-NCP with $\eta' = \frac{\log^2 n}{n}$

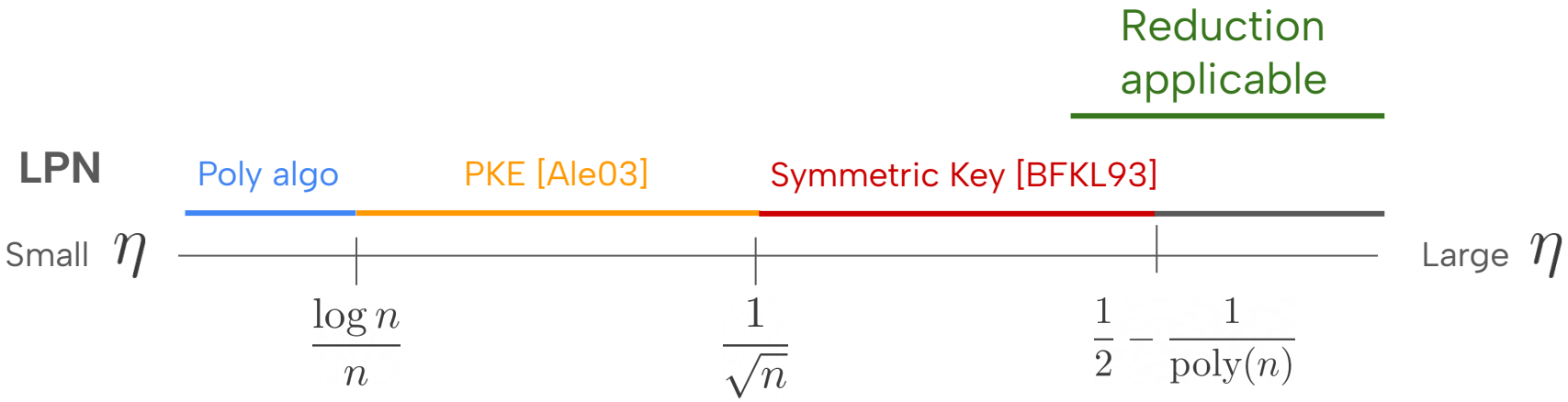
Bal-NCP is NCP with extra condition on matrix

# Worst-Case-to-Average-Case



# Reduction Limitations

- No worst-case hardness for low noise LPN
- Only show quasi-poly hardness for LPN at high noise



# Combined Worst-case Problem

For all $A \in \mathbb{F}_2^{m \times n}$ $s \in \mathbb{F}_2^n$ Solve either NCP or Dual Decision NCP

**NCP:**

Recover $s$ from $(A, As + e)$ $e \leftarrow Ber(\frac{1}{n^{2\alpha}})^m$

# Combined Worst-case Problem

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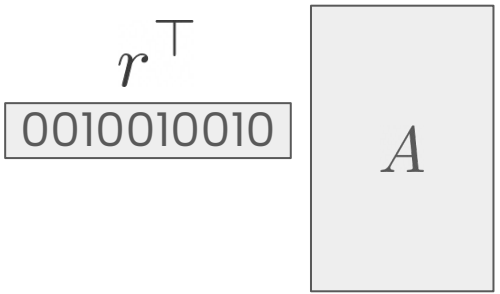
**NCP:**

Recover $s$ from $(A, As + e)$ $e \leftarrow Ber(\frac{1}{n^{2\alpha}})^m$

**Dual Decision NCP:**

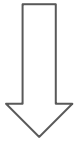
Distinguish $(A, r^\top A)$ from $(A, Ber(1/2)^n)$ $r \leftarrow$ Random length m vector with fixed weight $|n^a|$

Information Set Decoding: $2^{O(|r|)}$



# Main Theorem

If exists algorithm for LPN with noise rate $\frac{1}{n^{\alpha}}$



Exists two algorithms (**AlgS**, **AlgD**)

For all $A \in \mathbb{F}_2^{m \times n}$ $s \in \mathbb{F}_2^n$

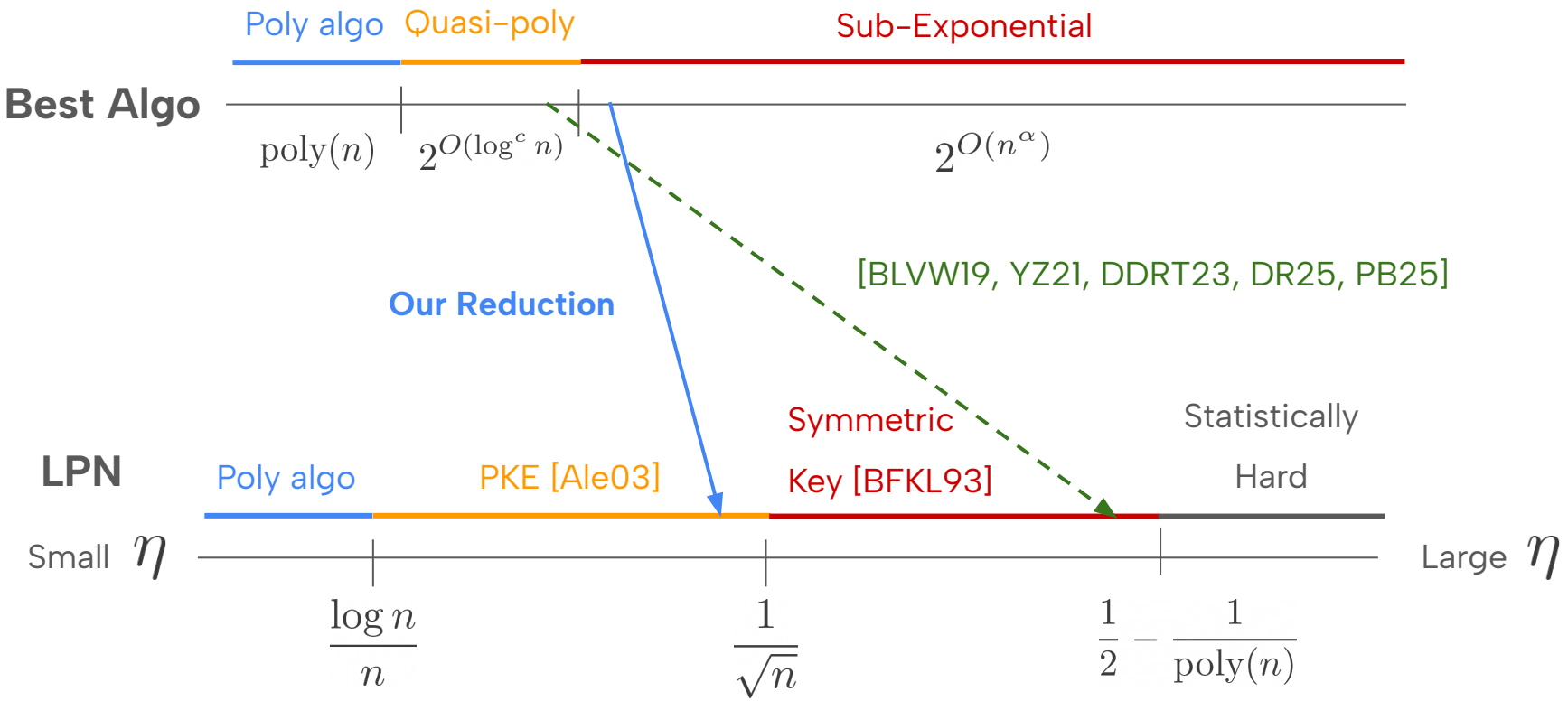
**Either** **AlgS** recover $s$ from $(A, As + e)$

**Or** **AlgD** distinguish $(A, r^{\top} A)$ from $(A, Ber(1/2)^n)$

$$e \leftarrow Ber(\frac{1}{n^{2\alpha}})^m$$

$r \leftarrow$ Random length m vector with fixed weight $|n^a|$

# Average-Case Hardness

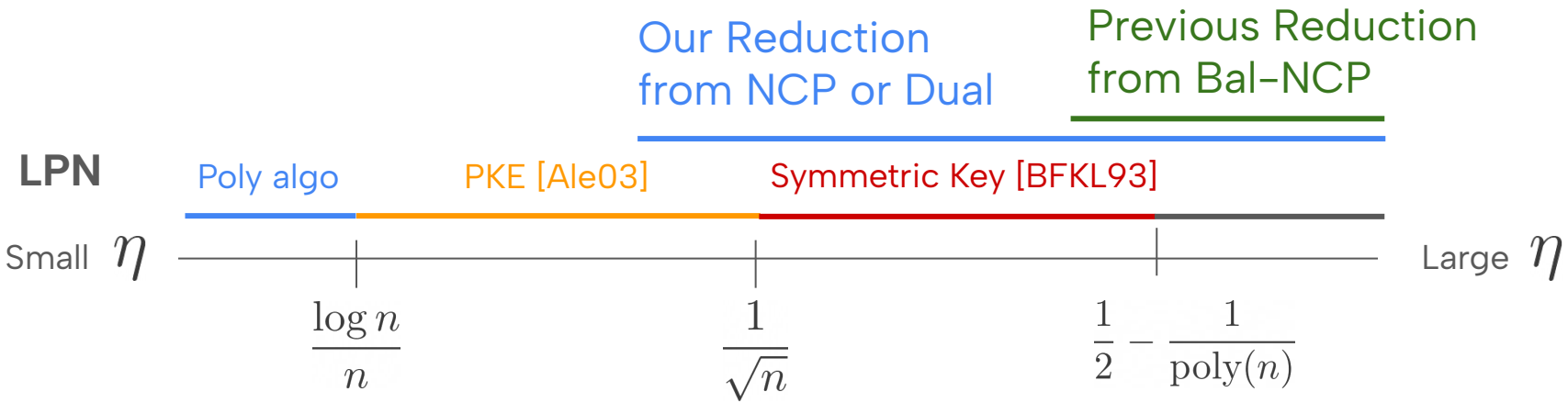


# Implication

If there exists at least one $A$ that **NCP** and **Dual Decision NCP** both hard

We obtain hardness for LPN with error rate $\frac{1}{n^{\alpha}}$

Such that it can be used to construct PKE [Ale03]



# Why noise increase so much?

Start with worst-case instance $A \in \mathbb{F}_2^{m \times n} \quad (A, As + e)$

Goal: Obtain average-case instance

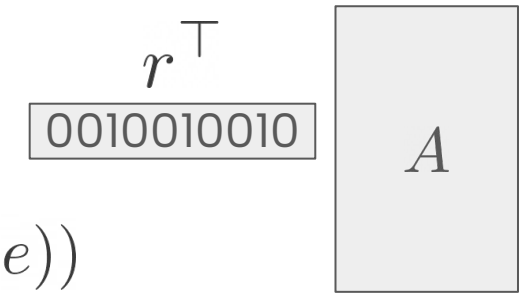
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**Reduction:** [BLVW19, YZ21, DDRT23, DR25, PB25]

1. Sample $r \leftarrow Ber(\tau)^m$
2. Obtain average-case sample $(r^\top A, r^\top (As + e))$



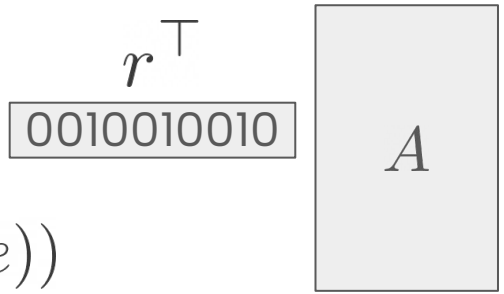
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**New Equation :** $r^\top A$ Statistically random if τ large , and A is well formed

**New Error :** $r^\top e$ If τ small, error rate small

# Our Proposal

**New Equation :** $r^\top A$ Computationally indistinguishable from random

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$$r^\top A \overset{comp}{\approx} U$$

Not clear ? ⇓

$$(r^\top A, r^\top(As+e)) \overset{comp}{\approx} (U, r^\top(As+e))$$

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The $r$ that we choose is **NOT** $r \leftarrow Ber(\tau)^m$

Rather, actual $r$ is random vector of fixed hamming weight

# Open Questions

- Extend the reduction to worst-case noise
- Understand the hardness of this joint worst-case problem